UNIVERSITI TUNKU ABDUL RAHMAN

ASSIGNMENT COVER SHEET

Course Code and Description	UECS1413 DATABASE SYSTEM FUNDAMENTALS
Lecturer Name	Dr.Sugumaran a/l Nallusamy
Assignment Title	Assignment 2
DECLARATION	
We declare that this is a group assignment and that no part of this submission has been copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part been written for us by another person.	

Programme	Student ID Numbers	Student Names	Practical Group
Software Engineering	2400219	Eric Gan Wei Hong	3
Software Engineering	2400258	Tham Yee Ting	3
Software Engineering	2305593	Chew Sin Yen	3
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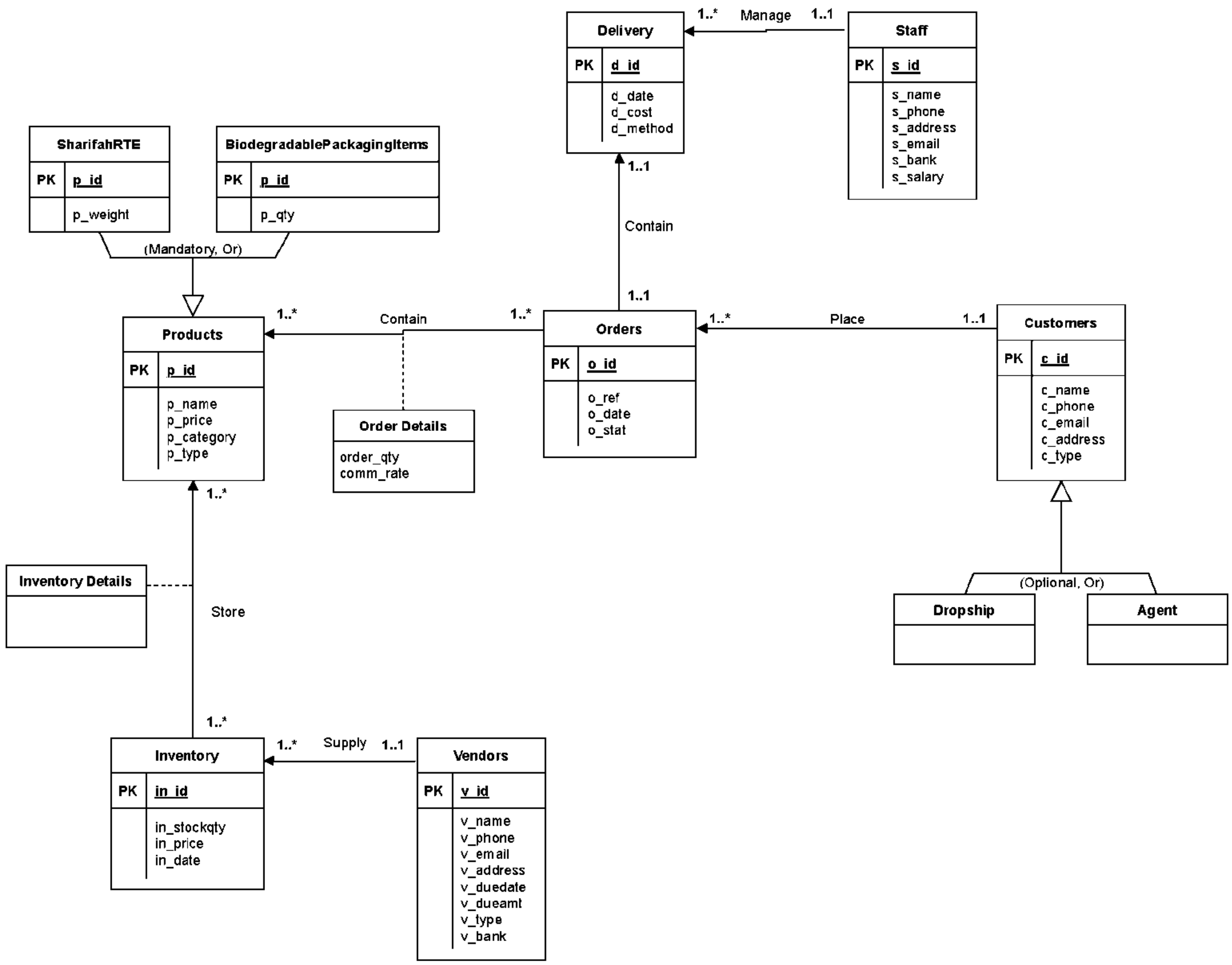
Description	Full Marks	Marks Obtained
Provide the correct or amended ERD diagram from the Programming Project Assignment I.	10	
*Remove features not compatible with the relational model *Derive relations for local logical data model.	40	
*Normalization Check The produce logical schema is expected to be in 3rdNF. Student may normalise their tables to produce the 3rdNF version, or (if the schema in 3rdNF, an assurance statement that the produced relational schema is in 3rdNF is essential).	5	
List the summary of relations with attributes, primary and foreign key; A **PK** may be identified by an underline (a common practice) and a FK by any other means as described by the student (e.g. double underline, italic style, different colour font, etc.)	15	
Draw the logical ERD diagram Provide the logical erd diagram with entity, relationship, attributes and multiplicity. Primary key and Foreign key also presented in the diagram	10	
Define data type logically	10	
Define the size logically	5	
All the columns define in the table above are listed in the Data Dictionary	5	
Total	100	

Amended ER Diagram



Incompatible Features Removed/Amended

Tables

Commissions (removed)

PK comm_id

FK o_id, p_id

comm_amount

comm_stat

Marketing Team Staff (removed)

PK mts_id

FK comm_id

mts_name

mts_phone

mts_email

mts_bank

mts_salary

Inventory Details (added)

PK in_id, p_id

FK in_id, p_id

Relations

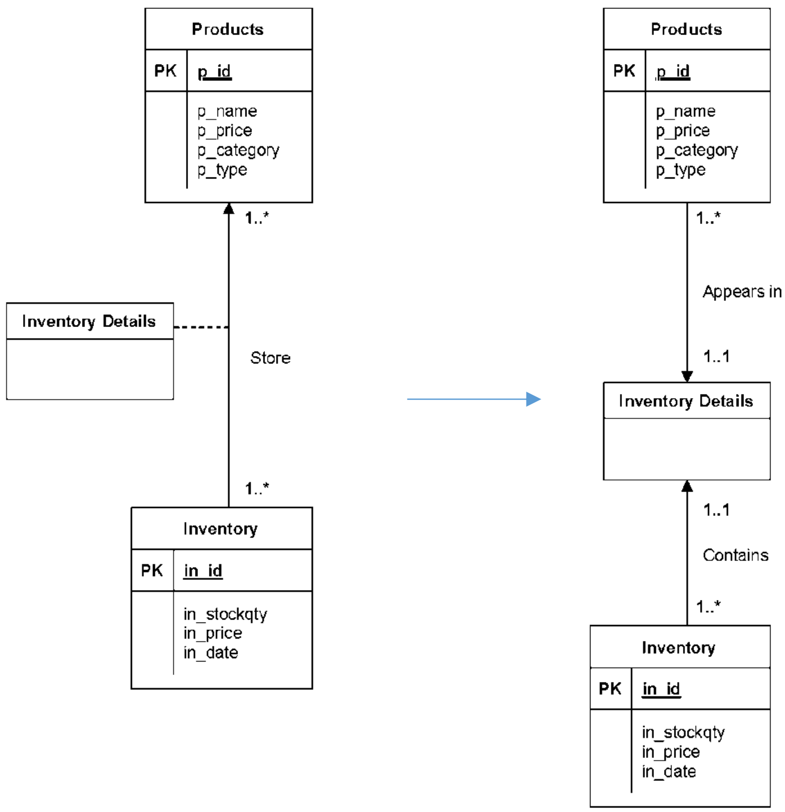
Staff 1..1 Communicate with 1..* → **Customers (removed)**

Order Details 1..1 Calculate 1..1 → **Commissions (removed)**

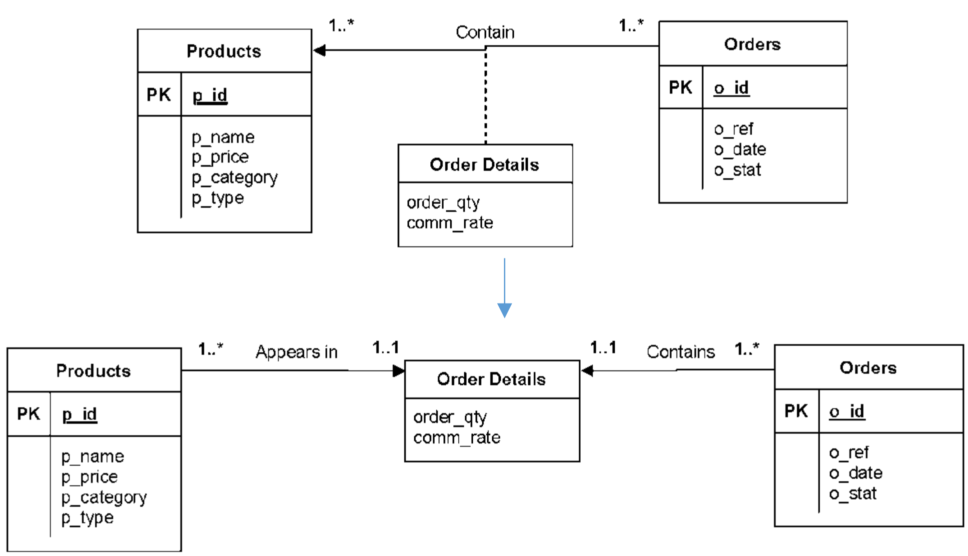
Commissions 1..* Given to 1..* → **Marketing Team Staff (removed)**

Remove *:* binary relationship

Products and Inventory



Products and Orders



<u>Derive Relations For Logical Data Model</u>

<u>1:1 Binary relationship type</u>

1. Post **o_id** into **Delivery** to model 1..1 **is delivered via** relationship.

Orders (**o_id**, *c_id,* o_ref, o_date, o_stat)

Primary Key o_id

Foreign Key *c_id* **references** Customers (c_id)

Delivery (**d_id**, *s_id, o_id,* d_date, d_cost, d_method)

Primary Key d_id

Foreign Key *o_id* **references** Orders (o_id)

Foreign Key *s_id* **references** Staff (s_id)

<u>1:* Binary relationship type</u>

1. Post **p_id** into **Order_Details** to model 1...* **appears in** relationship.

Products (**p_id,** p_name, p_price, p_category, p_type, p_weight, p_qty)

Primary Key p_id

Order_Details (*p_id, o_id*, order_qty, comm_rate)

Primary key o_id, p_id

Foreign key *p_id* **references** Products (p_id)

Foreign key *o_id* **references** Orders (o_id)

2. Post **o_id** into **Order_Details** to model 1...* **contains** relationship.

Orders (o_id, *c_id*, o_ref, o_date, o_stat)

Primary key o_id

Foreign key *c_id* **references** Customers (c_id)

Order_Details (***p_id, o_id,*** order_qty, comm_rate)

Primary key p_id, o_id

Foreign key *o_id* **references** Orders (o_id)

Foreign key *p_id* **references** Products (p_id)

3. Post **o_id** into **Delivery** to model 1..* **is delivered via** relationship.

Orders (**o_id**, *c_id*, o_ref, o_date, o_stat)

Primary Key o_id

Foreign key *c_id* **references** Customers (c_id)

Delivery (**d_id**, s_id, *o_id*, d_date, d_cost, d_method)

Primary Key d_id

Foreign Key *o_id* **references** Orders (o_id)

Foreign Key *s_id* **references** Staff (s_id)

4. Post **c_id** into **Orders** to model 1..* **places** relationship.

Customers (**c_id**, c_name, c_phone, c_email, c_address, c_type, dropship, agent)

Primary Key c_id

Orders (**o_id**, *c_id*, o_ref, o_date, o_stat)

Primary Key o_id

Foreign Key *c_id* **references** Customers (c_id)

5. Post **s_id** into **Delivery** to model 1..* **manages** relationship.

Staff (**s_id**, s_name, s_phone, s_address, s_email, s_bank, s_salary)

Primary Key s_id

Delivery (**d_id**, *s_id*, *o_id*, d_date, d_cost, d_method)

Primary Key d_id

Foreign Key *s_id* **references** Staff (s_id)

Foreign Key *o_id* **references** Orders (o_id)

6. Post **in_id** into **Inventory_Details** to model 1..* **contains** relationship.

Inventory (**in_id**, *v_id*, in_stockqty, in_price, in_date)

Primary Key in_id

Foreign Key *v_id* **references** Vendors (v_id)

Inventory_Details (***p_id, in_id***)

Primary Key (**p_id**, **in_id**)

Foreign Key *in_id* **references** Inventory (in_id)

Foreign Key *p_id* **references** Products (p_id)

7. Post **p_id** into **Inventory_Details** to model 1..* **appears in** relationship.

Products (**p_id**, p_name, p_price, p_category, p_type, p_weight, p_qty)

Primary Key p_id

Inventory_Details (***p_id, in_id***)

Primary Key (**p_id**, **in_id**)

Foreign Key *p_id* **references** Products (p_id)

Foreign Key *in_id* **references** Inventory (in_id)

8. Post **v_id** into **Inventory** to model 1..* **supplies** relationship.

Vendors (v_id, v_name, v_phone, v_email, v_address, v_dueDate, v_dueAmt, v_type, v_bank)

Primary Key v_id

Inventory (**in_id**, *v_id*, in_stockqty, in_price, in_date)

Primary Key in_id

Foreign Key *v_id* **references** Vendors (v_id)

Normalization Check

3NF of Inventory Management System

Since each attribute in the entities depends directly on the primary key without partial or transitive dependency, the database is already normalized into 3NF.

Examples:

- In Customers, c_name, c_phone, c_email, c_address, c_type are fully dependent on c_id only.
- In Order_Details, order_qty and comm_rate are dependent on the combination of o_id and p_id.
- In Vendors, v_name, v_phone, v_email, v_dueamount, etc., are dependent on v_id.

Thus, normalization from 2NF to 3NF has been completed.

Primary Keys:

- Products: p_id
- Orders: o_id
- Customers: c_id
- Vendors: v_id
- Inventory: in_id
- Staff: s_id
- Delivery: d_id

Composite Primary Keys:

- Order_Details: (o_id, p_id)
- Inventory_Details: (p_id, in_id)

Normalization Assurance Statement

All derived relational tables from the ERD have been normalized to Third Normal Form (3NF). Each table includes a clear and unique primary key. All non-key attributes are fully functionally dependent on the primary key, and no partial or transitive dependencies exist. This ensures the schema is free of redundancy and supports data integrity and consistency across the system.

Summary of relation

*Primary key is underlined and *foreign key* is *italicised*.

Customers (c_id, c_name, c_phone, c_email, c_address, c_type, *s_id*, dropship, agent)

Primary key: c_id

Foreign key: *s_id*

Vendors (v_id, v_name, v_phone, v_email, v_address, v_duedate, v_dueamt, v_type, v_bank)

Primary key: v_id

Products (p_id, p_name, p_price, p_category, p_type, p_weight, p_qty)

Primary key: p_id

Orders (o_id, o_ref, o_date, o_stat, *c_id*)

Primary key: o_id

Foreign key: *c_id*

Order_Details (*o_id, p_id*, order_qty, comm_rate, o_id, p_id)

Primary key: o_id, p_id

Foreign key: *o_id, p_id*

Inventory (in_id, in_stockqty, in_price, in_date, *v_id*, *p_id*)

Primary key: in_id

Foreign key: *v_id, p_id*

Inventory_Details (*in_id, p_id*)

Primary key: in_id, p_id

Foreign key: *in_id, p_id*

Staff (s_id, s_name, s_phone, s_address, s_email, s_bank, s_salary)

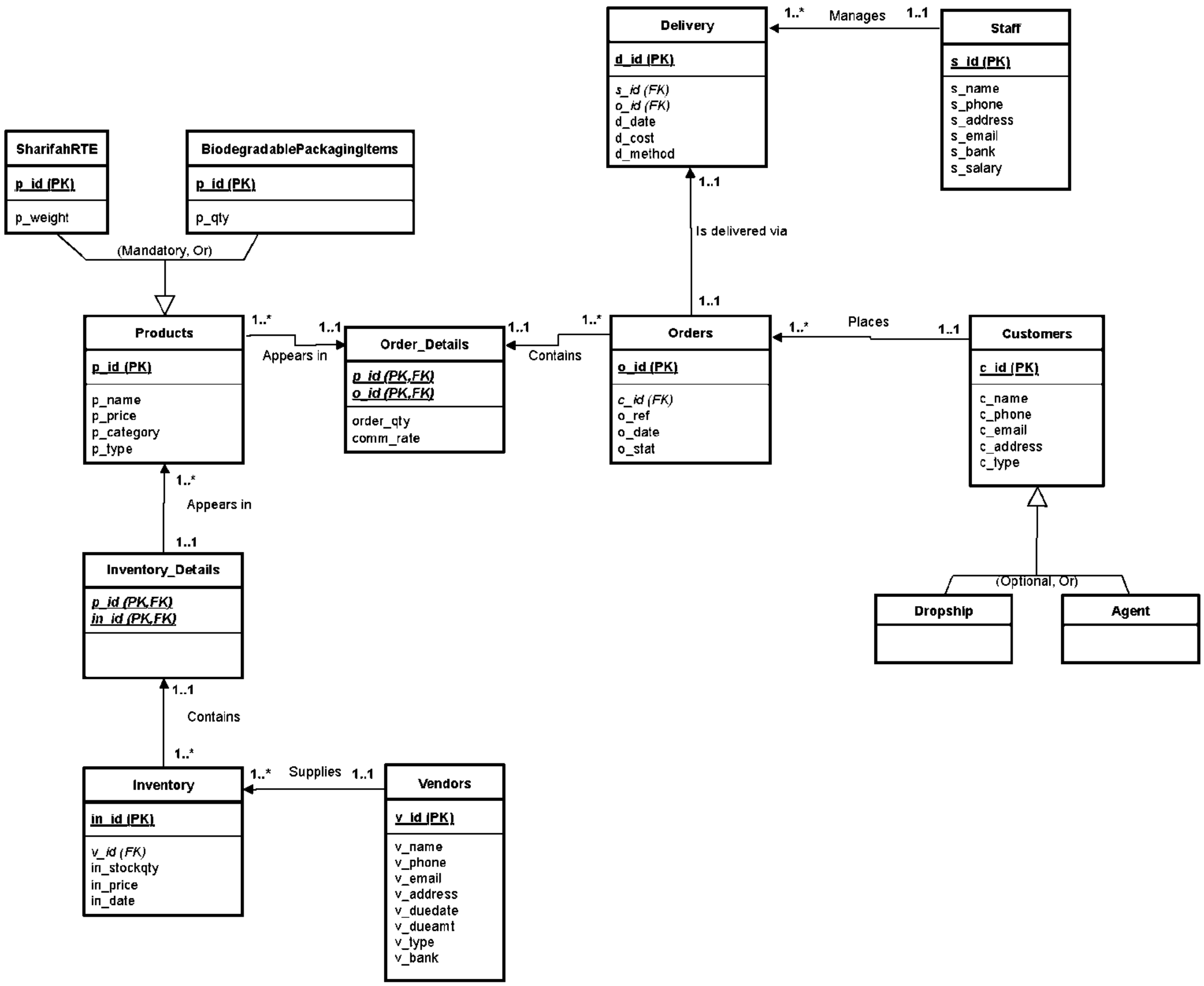
Primary key: s_id

Delivery (d_id, d_date, d_cost, d_method, *o_id, s_id*)

Primary key: d_id

Foreign key: *o_id, s_id*

Logical ER Diagram



Data Dictionary

1. **Table Name: Customers**

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK Referenced Table
c_id	Customer ID	Number	10	Yes	-	-
c_name	Customer Name	Text	40	-	-	-
c_phone	Phone Number	Number	15	-	-	-
c_email	Email Address	Text	30	-	-	-
c_address	Customer Address	Text	150	-	-	-
c_type	Customer Type	Text	30	-	-	-
dropship	Dropship	Text	30	-	-	-
agent	Agent	Text	30	-	-	-

2. **Table Name: Staff**

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK Referenced Table
s_id	Staff ID	Text	10	Yes	-	-
s_name	Staff Name	Text	40	-	-	-
s_phone	Phone Number	Number	15	-	-	-
s_email	Email Address	Text	30	-	-	-
s_address	Staff Address	Text	150	-	-	-
s_bank	Bank Details	Number	30	-	-	-
s_salary	Salary	Number	15	-	-	-

3. **Table Name: Products**

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK Referenced Table
p_id	Product ID	Text	10	Yes	-	-
p_name	Product Name	Text	40	-	-	-
p_price	Price	Number	15	-	-	-
p_category	Product Category	Text	20	-	-	-
p_type	Product Type	Text	30	-	-	-
p_weight	Weight of Product	Number	20	-	-	-
p_qty	Quantity of Product	Number	20	-	-	-

4. **Table Name: Inventory**

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK Referenced Table
in_id	Inventory ID	Text	10	Yes	-	-
in_price	Inventory Price	Number	15	-	-	-
in_stockqty	Stock Quantity	Number	20	-	-	-
in_date	Inventory Date	Date	20	-	-	-
v_id	Vendor ID	Text	10	-	Yes	Vendors

5. **Table Name: Inventory Details**

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK Referenced Table
p_id	Product ID	Text	10	Yes (part)	Yes	Product
in_id	Inventory ID	Text	10	Yes (part)	Yes	Inventory

6. **Table Name: Vendors**

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK Referenced Table
v_id	Vendor ID	Text	10	Yes	-	-
v_name	Vendor Name	Text	30	-	-	-
v_phone	Phone Number	Number	15	-	-	-
v_email	Email Address	Text	40	-	-	-
v_address	Address	Text	50	-	-	-
v_duedate	Payment Due Date	Date	20	-	-	-
v_dueamt	Amount Due Date	Date	20	-	-	-
v_type	Vendor Type	Text	30	-	-	-
v_bank	Bank Details	Number	30	-	-	-

7. **Table Name: Orders**

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK Referenced Table
o_id	Order ID	Text	10	Yes	-	-
o_ref	Order Reference	Text	30	-	-	-
o_date	Order Date	Date	20	-	-	-
o_stat	Order Status	Text	20	-	-	-
c_id	Customer ID	Text	10	-	Yes	Customers

8. **Table Name: Order_Details**

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK Referenced Table
o_id	Order ID	Text	10	Yes (part)	Yes	Orders
p_id	Product ID	Text	10	Yes (part)	Yes	Products
order_qty	Quantity Ordered	Number	10	-	-	-
comm_rate	Commission Rate	Number		-	-	-

9. **Table Name: Delivery**

Column Name	Description	Data Type	Size	Primary Key?	Foreign Key?	FK Referenced Table
d_id	Delivery ID	Text	10	Yes	-	-
d_date	Delivery Date	Date	20	-	-	-
d_cost	Delivery Cost	Number	10	-	-	-
d_method	Delivery Method	Text	30	-	-	-
o_id	Order ID	Text	10	-	Yes	Orders
s_id	Staff ID	Text	10	-	Yes	Staff